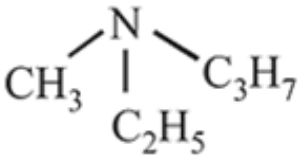
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Which statement is true regarding the following structure?



(A) It is a chiral molecule

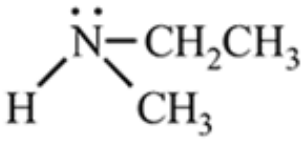
(B) It exists in two resolvable optically active forms

(C) Both (a) and (b)

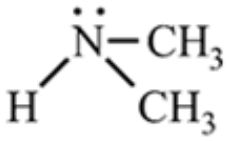
(D) Neither (a) nor (b)

Q.2

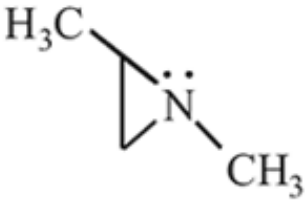
Which of the following amines can be resolved into two enantiomers?



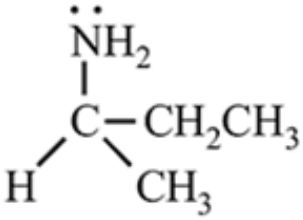
I



II



III



IV

(A) I,IV

(B) I, II

(C) I, III, IV

(D) III, IV

Q.3

Number of stereo isomers existing for



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(A) 1

(B) 2

(C) 3

(D) 4

Q.4

The optically inactive compound from the following is :

(A) 2 - chloropropanal

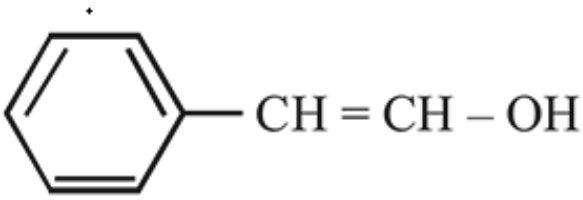
(B) 2 -chlorobutane

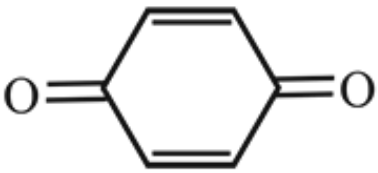
(C) 2 - chloropentane

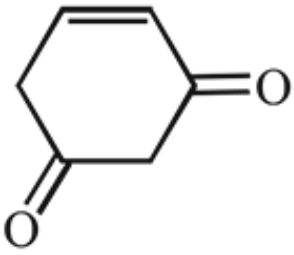
(D) 2 - chloro - 2- methylbutane

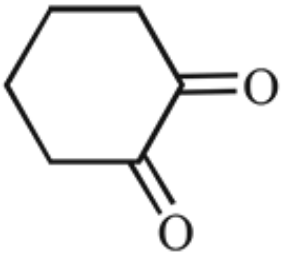
Q.5

Tautomerism is exhibited by-

(1) 

(2) 

(3) 

(4) 

(A) $(1), (3)$ and (4)

(B) $(2), (3)$, and (4)

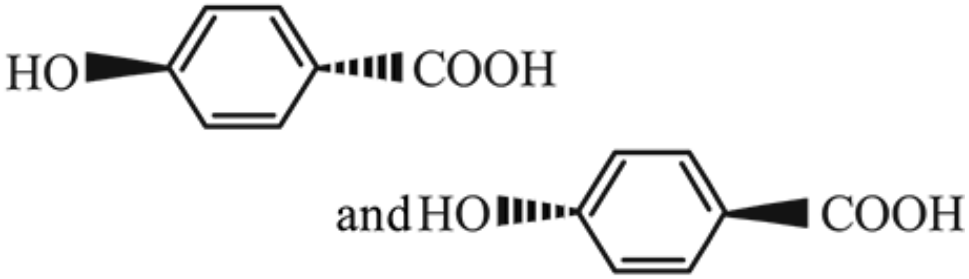
(C) (1) (2) and (4)

(D) None of these

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Q.6

The two compounds given below are



(A) identical

(B) enantiomers

(C) diastereo isomers

(D) regiomers

Q.7

The number of structural isomers for C_6H_{14} is:

(A) 4

(B) 3

(C) 6

(D) 5

Q.8

Which of the following pairs of compounds are positional isomers?

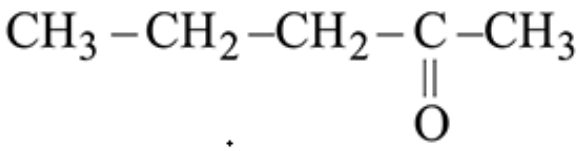
(A) $CH_3-CH_2-CH_2-\overset{\overset{O}{\|}}{C}-CH_3$

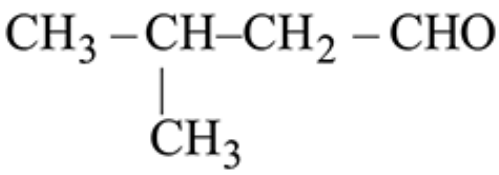
and $CH_3-CH_2-\overset{\overset{O}{\|}}{C}-CH_2-CH_3$

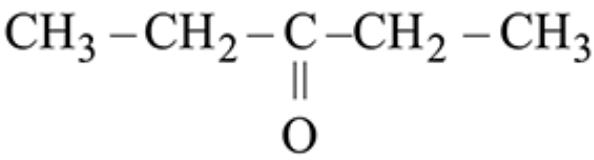
(B) $CH_3-CH_2-CH_2-CH_2-CHO$

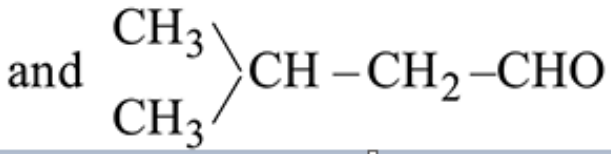
and $CH_3-CH_2-CH_2-\overset{\overset{O}{\|}}{C}-CH_3$

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(C) 

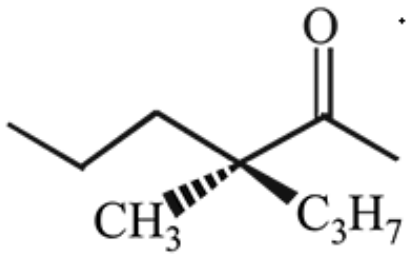
and 

(D) 

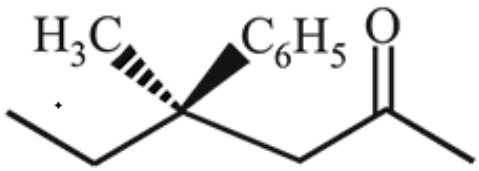
and 

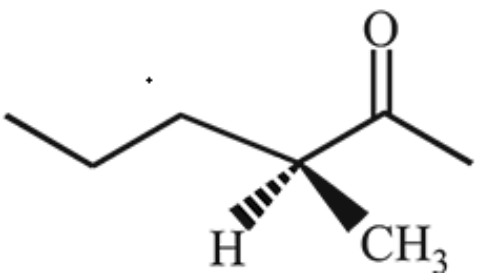
Q.9

Which of the following optically active compounds racemizes in dil. KOH/CH_3OH solution?

(A) 

(B) 

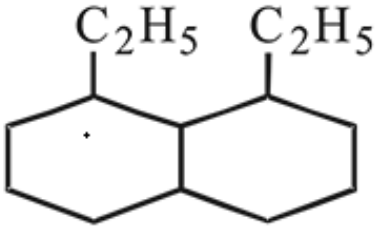
(C) 

(D) 

Q.10

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The correct number of chiral centres in



(A) 1

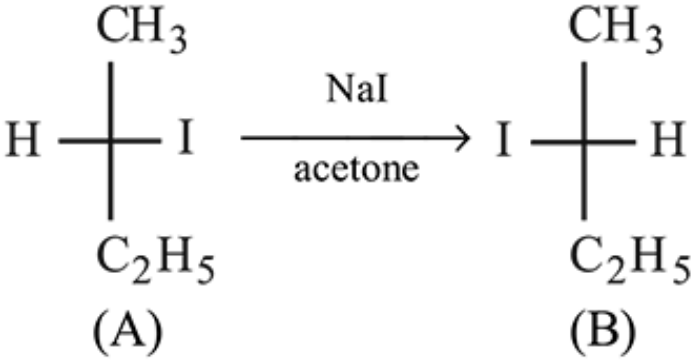
(B) 3

(C) 4

(D) 2

Q.11

An optically active compound (A) is treated with NaI/acetone (B) is obtained



(A) (B)

(A) optically active

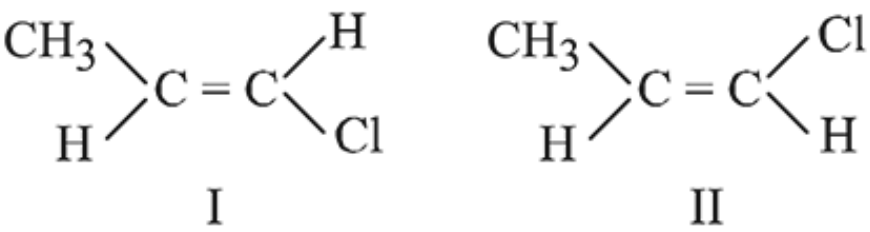
(B) optically inactive and racemic form

(C) optically inactive

(D) None of these

Q.12

Which of the following is correct set of physical properties of the geometrical isomers?



I II

Dipole moment	B.P.	M.P.	Stability

(A) I > II

I > II II > I I > II

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(B) $\mathrm{II} > \mathrm{I}$
$\mathrm{II} > \mathrm{I} \quad \mathrm{II} > \mathrm{I} \quad \mathrm{II} > \mathrm{I}$

(C) $\mathrm{I} > \mathrm{II}$
$\mathrm{I} > \mathrm{II} \quad \mathrm{I} > \mathrm{II} \quad \mathrm{I} > \mathrm{II}$

(D) $\mathrm{II} > \mathrm{I} \quad \mathrm{II} > \mathrm{I} \quad \mathrm{I} > \mathrm{II} \quad \mathrm{I} > \mathrm{II}$

Q.13

A solution of $(+) - 2$ - chloro -2 - phenylethane in toluene racemises slowly in the presence of small amounts of $\mathrm{SbCl_5}$ due to the formation of

(A) Carbanion

(B) Carbene

(C) Free radical

(D) Carbocation

Q.14

Which one of the following acids does not exhibit optical isomerism?

(A) Lactic acid

(B) Tartaric acid

(C) Maleic acid

(D) α -amino acids

Q.15

How many cyclic structures are possible for
$\mathrm{C_4H_6}$?

(A) 3

(B) 5

(C) 6

(D) 4

Q.16

Maleic acid and fumaric acids are

(A) Chain isomers

(B) Functional isomers

(C) Tautomers

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(D) Geometrical isomers

Q.17

Which one of the following conformations of cyclohexane is chiral?

(A) Boat

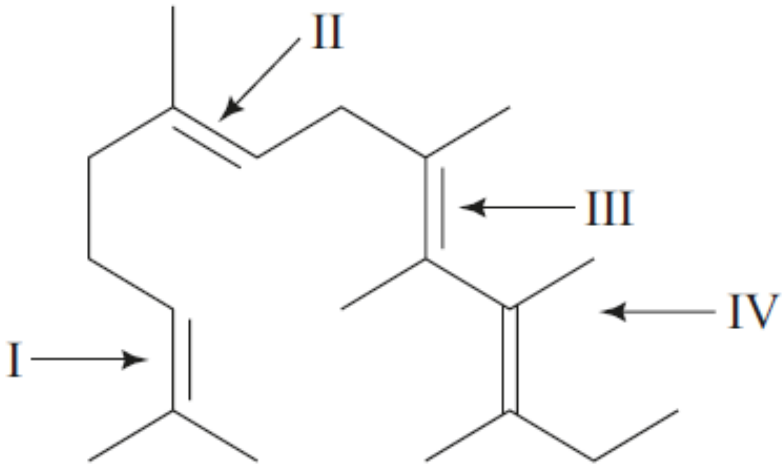
(B) Twist boat

(C) Rigid

(D) Chair

Q.18

In the following structure, the double bonds are marked as I, II, III and IV. Geometrical isomerism is not present at site (s) :



(A) I and II

(B) III

(C) I

(D) III and IV

Q.19

What is the increasing order of stability of the three main conformations (i.e., *Eclipse, Anti, Gauche*) of 2-fluoroethanol ?

(A) *Eclipse, Gauche, Anti*

(B) *Gauche, Eclipse, Anti*

(C) *Eclipse, Anti, Gauche*

(D) *Anti, Gauche, Eclipse*

Q.20

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In open chain structure, how many chiral carbon atoms in D-(-)-Ribose ?

(A) 3

(B) 4

(C) 2

(D) 5

Answers & Solutions

Q.1 Answer:

It is a chiral molecule

Solution:

In amines, N is sp^3 hybridised and thus has pyramidal shape. In the given structure, since the three alkyl groups are different, and the fourth corner of the pyramid is occupied by lone pair of electrons, the molecule is chiral. However, the two enantiomers of the amine are not resolvable because of their rapid interconversion through a transition state having planar structure (sp^2 hybridised nitrogen)

Q.2 Answer:

III, IV

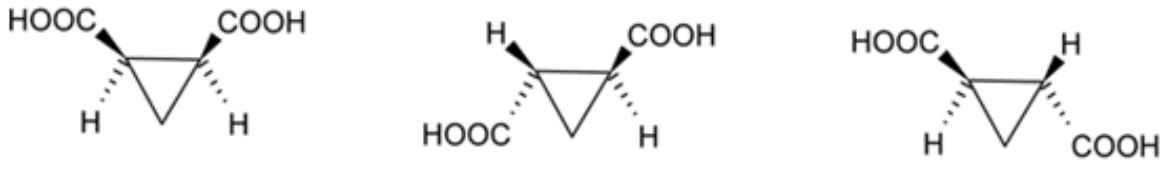
Solution:

Most of the simple amines those having smaller alkyl (groups) can't be resolved, although they may contain three different alkyl groups.

Q.3 Answer:

3

Solution:



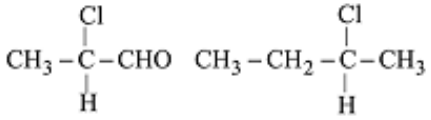
Q.4 Answer:

2 - chloro - 2- methylbutane

Solution:

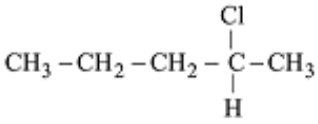
The optically inactive compound must contains achiral carbon atom(s). Option (d) contains achiral carbon atom

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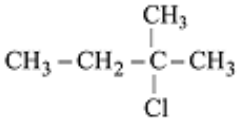


2- chloropropanol

2-chlorobutane (chiral)



2-chloropentane
(chiral)

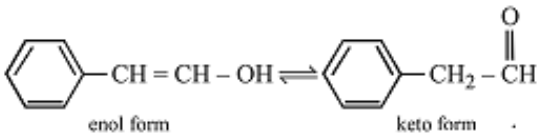


2-chloro-2-methylbutane
(achiral)

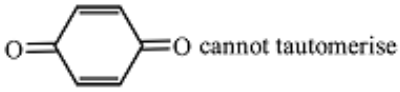
Q.5 Answer:

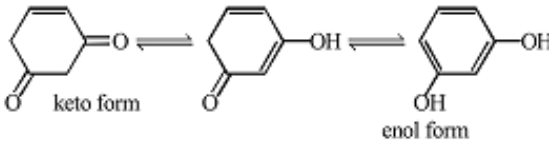
(1), (3) and (4)

Solution:

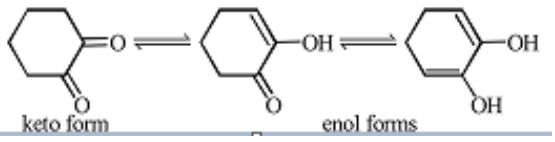
(a) 

enol form keto form

(b)  cannot tautomerise

(c) 

keto form enol form

(d) 

keto form enol forms

Q.6 Answer:

identical

Solution:

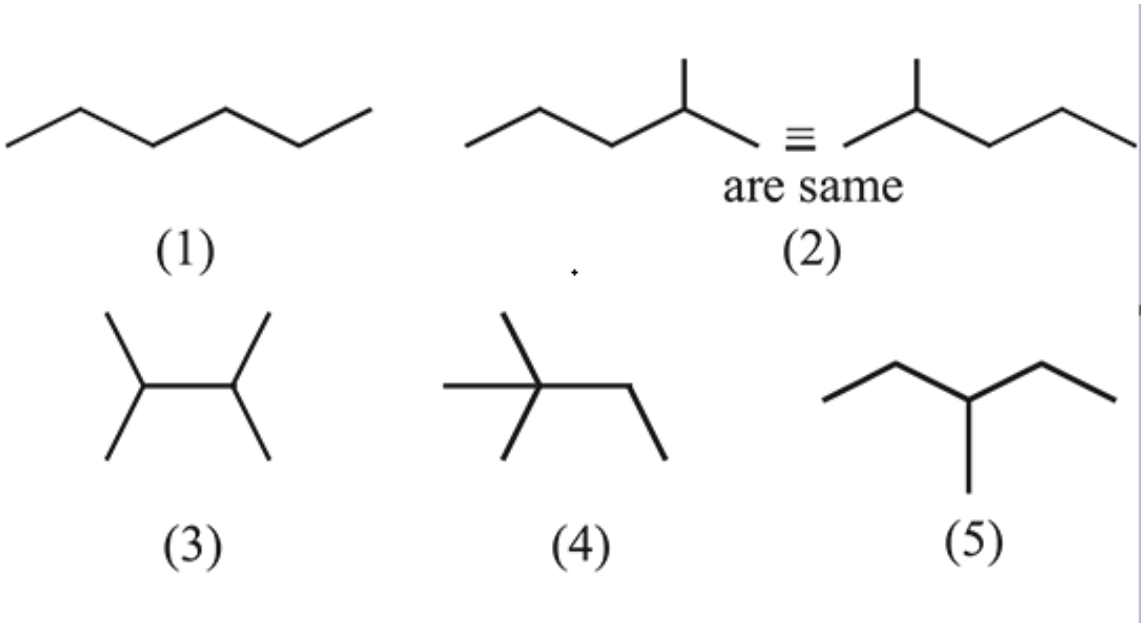
The given compounds are identical

Q.7 Answer:

5

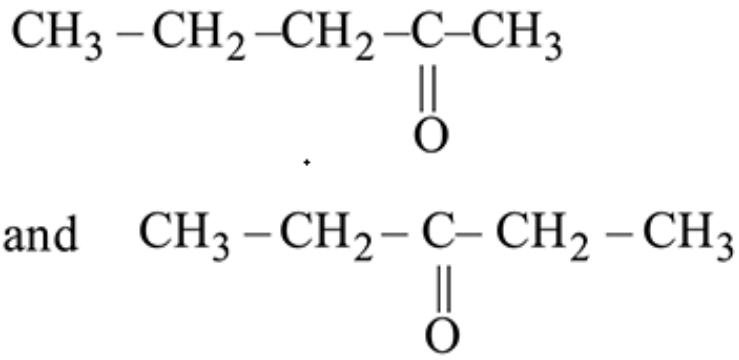
Solution:

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Total 5 structural isomers

Q.8 Answer:



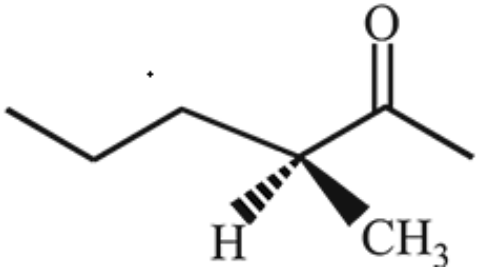
and

Solution:

Pentan-2-one and pentan-3-one are position isomers.

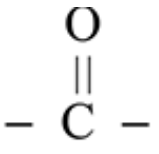
(b), (c), (d) contain different compounds aldehyde and ketones

Q.9 Answer:



Solution:

The compounds containing



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group adjacent to asymmetric carbon atom carrying a hydrogen easily racemise.

Q.10 Answer:

2

Solution:

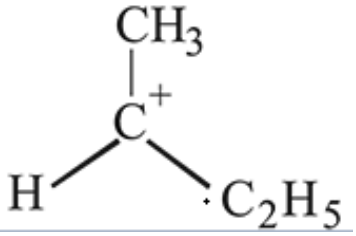
The molecule contains two similar chiral centres.

Q.11 Answer:

optically inactive and racemic form

Solution:

The reaction proceeds through planar leading to racemisation.



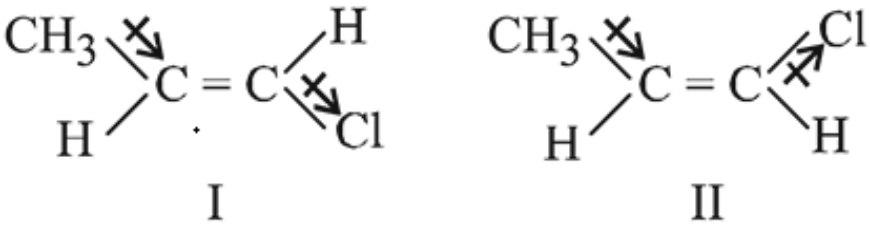
Q.12 Answer:

I > II

I > II I > II I > II

Solution:

In compounds



I II

I has more dipole moment than II, hence its boiling point will be higher. Melting point depends on symmetry, therefore I has higher melting point than II. Steric crowding in II is more than in I therefore I is more stable than II

Q.13 Answer:

Carbocation

Solution:

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$SbCl_5$ pulls Cl^- to form $SbCl_6^-$ leaving behind planar $C_6H_5 - CH^+ - CH_3$ carbonium ion. It can be attacked from either side leading to racemic mixture.

Q.14 Answer:

Maleic acid

Solution:

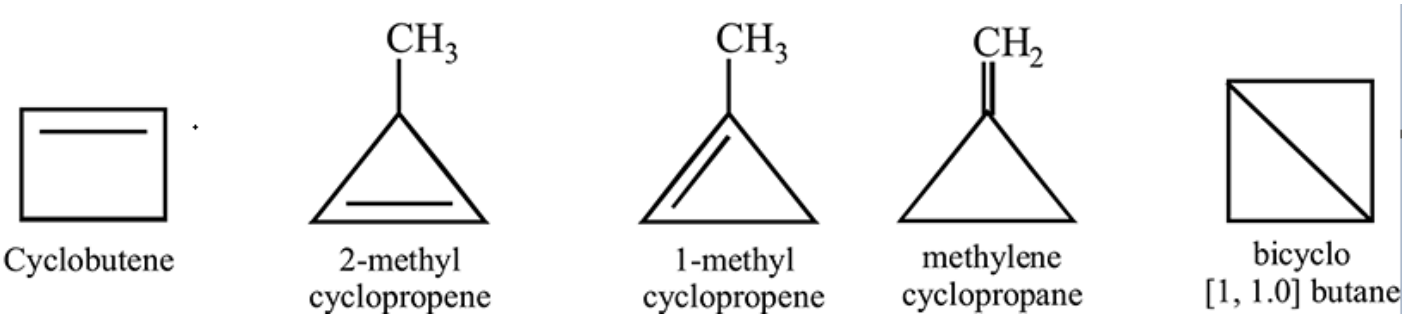
Optical isomers contain an asymmetric (chiral) carbon atom (a carbon atom attached to four different atoms or groups). Therefore all except maleic acid exhibit optical isomerism.

Q.15 Answer:

5

Solution:

Five cyclic structures are possible for C_4H_6. These are as following:



Cyclobutene | 2-methyl cyclopropene | 1-methyl cyclopropene | methylene cyclopropane | bicyclo [1, 1.0] butane

Q.16 Answer:

Geometrical isomers

Solution:

Maleic acid and fumaric acids are geometrical isomers.



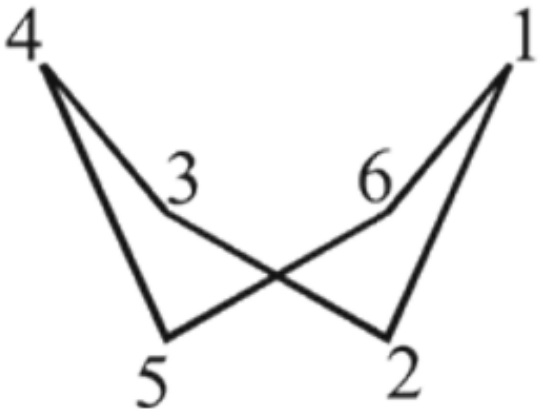
Q.17 Answer:

Twist boat

Solution:

Chiral conformation will not have plane of symmetry. Since twist boat does not have plane of symmetry it is chiral.

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Q.18 Answer:

I

Solution:

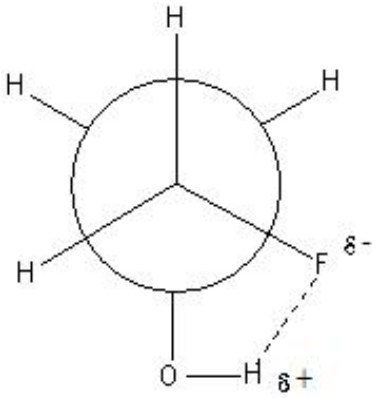
The site I contains two identical CH_3 groups attached to double bond. This site will not show geometrical isomerism.

Q.19 Answer:

Eclipse, Anti, Gauche

Solution:

$HO - CH_2 - CH_2 - F$



Gauche **conformation is comparatively more stable due to hydrogen linkage in between F and H (at O-atom), hence order is** *Eclipse, Anti* **(staggered),** *Gauche***.**

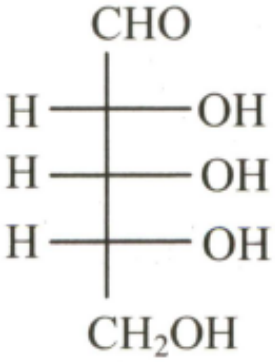
Q.20 Answer:

3

Solution:

Structure of D**-(-)-Ribose is:**

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The number of chiral carbon atoms is 3

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